

Angle dependence of Brillouin scattering intensity in liquids: supplement

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Appendix 1 Gaussian envelope

As the name implies, a Virtually Imaged Phased Array (VIPA) can be viewed as a series of virtual sources that interfere with each other, similar to a phased array [1]. The phase difference between these sources is determined by both the input and output angles, which is also frequency dependent. As a result, the direction of the output beam varies with frequency, akin to a diffraction grating. Furthermore, due to the intensity distribution of each light source, there is an intensity modulation in the interference pattern.

According to the analysis by Zhu et al. [2], the intensity distribution in the focal plane after the VIPA is:

$$I_F(u) \propto \tilde{U}_0^2(u) \frac{1}{1 + P \sin^2(k\xi)} \quad \#(S1)$$

where u is the coordinate in the focal plane, k is the light wave vector, P is a parameter related to the reflectivities of the VIPA, and ξ is the phase difference. The first term $\tilde{U}_0^2(u)$ represents the Fourier transform of the light intensity distribution at the front plane of the VIPA, which induces the diffraction envelope. For light output from a single-mode fiber, this envelope is Gaussian.

To accurately measure the intensity of each frequency component, it is necessary to exclude the influence of this envelope. The surface of the vial was placed at the focal plane of the lens, and the intense reflected laser signal was collected. By analyzing different diffraction orders of this signal, we fitted the envelope using a Gaussian function.

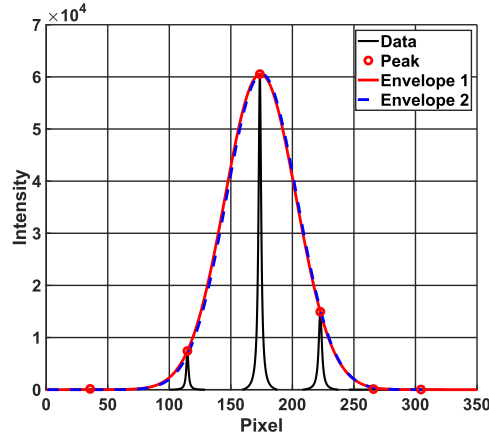


Fig. S1. Gaussian envelope induced by the VIPA. Envelope 1 is fitted to the peaks of multiple diffraction orders from the laser source signal, while envelope 2 is derived by scanning the laser source frequency.

To validate the feasibility of the method, we compared the results with those obtained by scanning the frequency of the laser. As shown in Fig. S1, the intensities in different diffraction orders of the laser signal are modulated and exhibit significant variation. Two envelopes were fitted based on the peak positions of each diffraction order, and only a slight difference between them was observed. It is important to note that we use a fused-silica VIPA, which is sensitive to temperature fluctuations, so the envelope position is recalculated at the start of each experiment. Additionally, although the laser frequency is locked during the experiment, occasional fluctuations may still occur, potentially causing deviations in the intensity

calculation. To minimize the envelope's influence, the tilt angle of the VIPA is slightly adjusted to ensure that the diffraction order of interest remains centered within the Gaussian envelope. Upon reviewing the data shown in Fig. 2, we found that the mean value of the Landau-Placzek ratios is only 13% higher before correction, which falls within the acceptable tolerance range.

Appendix 2 Data Collection and Processing

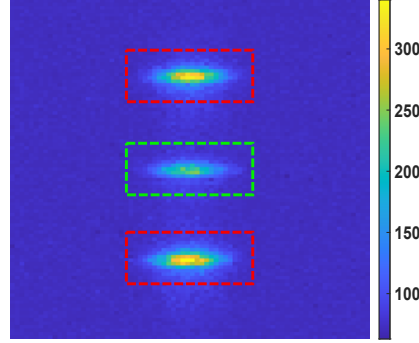


Fig. S2. Characteristic Rayleigh-Brillouin spectrum in liquids captured by the camera (raw data). Three identical regions of interest are selected, with the green region representing the Rayleigh component and the red regions representing the Brillouin components.

A characteristic Rayleigh-Brillouin spectrum in liquids is shown in Fig. S2, with weaker Rayleigh scattering at the center and non-uniform background noise from the camera. In each experiment, we measure the scattering signals with the sample and the background noise without the sample. The background noise, primarily from the camera, is subtracted from the scattering signals. The signals are then corrected by compensating for the intensity modulation introduced by the envelope, which is calculated in advance. Three identical regions of interest are selected from the corrected two-dimensional image, corresponding to the Stokes, Anti-Stokes, and Rayleigh signals. By integrating the pixel values in each region, we obtain the intensities for the Rayleigh and Brillouin signals, and we apply the "three-sigma rule" to filter out dust scattering. The Landau-Placzek ratio is then calculated, with the mean value set as the final result.

References

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